# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Second Semester of M. Tech. (CAD/CAM) Examination**

**Nov 2013**

**ME 709.01 Advance Engineering Optimization Techniques**

**Date: 26.12.2013, Thursday** **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

**Q - 1 (a)** Find out the best work-tool combinations in terms of machinability for the data given in Table 1 using TOPSIS methods. The attributes considered are tool wear rate (TW), specific energy consumed (SE), and processed surface roughness (SR).The weights for the attributes are WTW = 0.7306; WSE = 0.1884; WSR = 0.0810 resp. **[09]**

Table:1 Objective data of the machinability attributes

| Work-Tool Combination | Tool wear rate (m/min) | Specific energy consumed (N) | Surface Roughness (μm) |
|---|---|---|---|
| 1 | 0.061 | 219.74 | 5.8 |
| 2 | 0.028 | 761.46 | 5.8 |
| 3 | 0.034 | 1593.48 | 5.8 |
| 4 | 0.013 | 2849.15 | 6.2 |

1: TiAl6V4-P20; 2: TiMo32-P20; 3: TiAl5Fe2.5-P20; 4: TiAl6V4-P20 (TiN);

**Q - 2 (a)** Find maxima and minima, if any, of the function **[07]**

$$f(x) = 4x^3 - 18x^2 + 27x - 7$$

**(b)** Determine whether the following function is concave , convex or neither **[07]**

$$f(x) = x_1^2 + 3x_1x_2 + 2x_2^2$$

**OR**

**Q - 2 (a)** Solve the following function using Lagrange's Multiplier **[07]**

$$\begin{aligned} &\textit{Maximise} \quad Z = 5x_1 + x_2 - (x_1 - x_2)^2, \\ &\textit{Subject to} \quad x_1 + x_2 = 4, \\ &\qquad\qquad x_1, x_2 \geq 0 \end{aligned}$$

(b) Determine whether the following function is concave , convex or neither [07]

$f(x) = x_1 x_2$

**Q - 3 Attempt short notes on Any TWO** [12]

(a) Fuzzy Membership functions

(b) Components of Genetic Algorithm

(c) Artificial Neural Network Architecture

## SECTION – II

Q - 4 (a) Define Local and Global Optimum. Also discuss about inflexion point. [04]

(b) Define Engineering Optimization and enlist the applications. [04]

Q - 5 (a) Enlist different elimination methods based on final interval of uncertainty [04]

(b) Find the minimum of $f(x) = x^5 - 5x^3 - 20x + 5$ by exhaustive search in the interval of (0, 5). [08]

**OR**

Q - 5 (a) Define unimodal function? Classify methods used to approximate solution of such a function [04]

(b) Write algorithm for approximation solution of a unimodal function with a Fibonacci method. [08]

**Q - 6 Attempt the following (Any three)** [15]

(a) C Program on Golden Section Method

(b) C Program on Newton Raphson's Method

(c) Algorithm of Powell's Method

(d) C Program on Unrestricted Search Method

(e) Algorithm on Steepest Decent Method

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